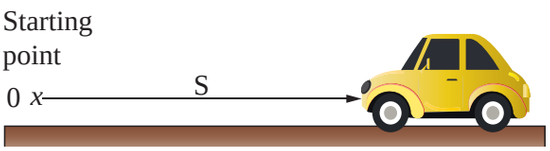
2.4 Distance – Time Graphs

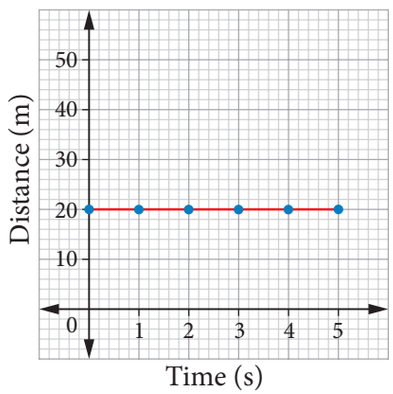
A car travelling along a straight line away from the starting point O is shown in the figure. The distance of the car is measured for every second. The distance and time are recorded and a graph is plotted using the data. The results for four possible journeys are shown below.

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a. Car at rest

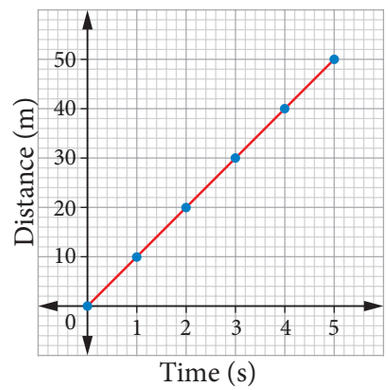
Time (s)	0	1	2	3	4	5
Distance (m)	20	20	20	20	20	20



The graph has zero gradient. i.e. the distance is constant for every second. Thus, the car is at rest.

b. Car travelling at uniform speed of 10ms^{-1}

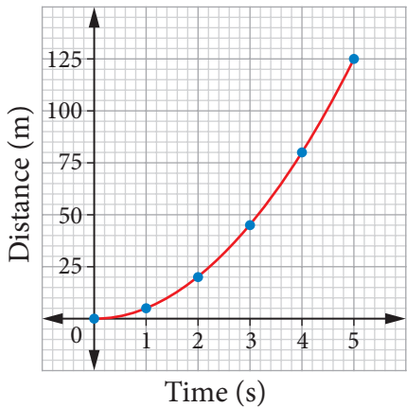
Time (s)	0	1	2	3	4	5
Distance (m)	0	10	20	30	40	50



The graph has constant gradient. The distance increases 10 m in every second. Thus, the car moves with uniform speed.

c. Car travelling at increasing speed

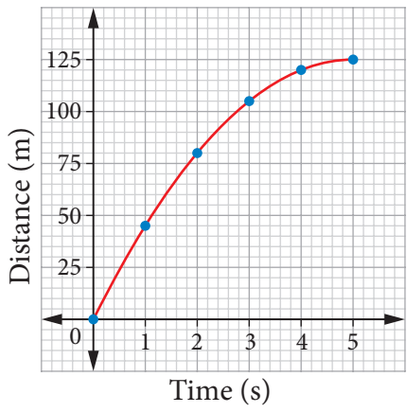
Time (s)	0	1	2	3	4	5
Distance (m)	0	5	20	45	80	125



The graph has an increasing gradient, i.e. That is, the speed increases.

d. Car travelling at decreasing speed

Time (s)	0	1	2	3	4	5
Distance (m)	0	45	80	105	120	125



The graph has a decreasing gradient. That is, the speed decreases.

2.5 Speed – Time Graphs

Let us consider a bus travelling from Thanjavur to Trichy. The speed of the bus is measured for every second. The speed and time are recorded and a graph is plotted using the data. It is known as speed-time graph. The results for four possible journeys are shown.

VII Std Science Term-1 EM Unit 2.indd 20 24-03-2020 14:09:15